Back

Time to Practice: Clustering Overview
Graded Assignment • 15 min

Due Oct 19, 11:59 PM IST

Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item →

1. Which of the following scenarios could *not* be considered a type of clustering problem? 1 / 1 point

- Grouping similar customers on a web site according to their shopping behavior
- Compressing data by replacing data points with the centroids of identified groups
- Predicting the topic of each web page in a set of search results based on their content
- Inferring missing attributes of a user's profile based on nearby users with similar profiles

Nice work

Correct: This is a supervised learning problem (classification), not a clustering problem.

2. What key criteria define a 'good' clustering? 1 / 1 point

- Low similarity of points within a cluster; each cluster has high similarity with points in other clusters
- High similarity of points within a cluster; each cluster well-separated from other clusters
- High variance within a cluster; low variance across clusters
- Large dissimilarity of points within a cluster; Large similarity of points that are in different clusters.

Nice work

Correct: these are the two main conditions clustering algorithms attempt to satisfy.

3. What is the first step in a typical cluster analysis? 1 / 1 point

- Apply clustering quality and validity measures
- Choose an appropriate data representation
- Choose the number of clusters
- Select a distance measure

Nice work

Correct. This is always a crucial way to being the clustering analysis: no other steps can proceed without it.

4. Which choice below is *not* one of the three desirable properties a clustering should try to satisfy, as discussed in Jon Kleinberg's 'clustering impossibility' paper? 1 / 1 point

- Local neighborhood simplicity
- Scale-invariance
- Richness
- Consistency

Nice work

This is not one of the three clustering properties.

5. Which of the following facts is *incorrect* when considering the difference between 'hard' and 'soft' clustering approaches? 1 / 1 point

- 'Hard' clustering is sometimes called 'partitional' clustering.
- 'Hard' clustering allocates each object to exactly 1 cluster
- 'Soft' clustering allows each object to belong to multiple clusters
- 'Hard' clustering defines a zone of required separation between clusters

Nice work

This definition is incorrect.

6. Which of the following clustering facts is *incorrect*? 1 / 1 point

- Hierarchical clustering creates a set of nested clusters organized as a tree
- Clustering can be used to handle missing data
- It is always possible to define a 'best' clustering of a given dataset
- According to Kleinberg's impossibility theorem, there is no clustering function that can satisfy those three stated desirable properties all at the same time.

Nice work

This is incorrect. There can be ambiguous situations where different clusterings may be considered equally effective.